

Requirements & Business Rules Specification

Chemical Injection Equipment Management System (CIEMS)

Chemical Injection Equipment Management System (CIEMS) | Business Analyst Challenge | Version 1.0

1. Introduction

1.1 Purpose

This document specifies the functional requirements, non-functional requirements, and business rules of CIEMS, a web system used to register chemical tanks, wells, and pumps, associate tanks to wells, and monitor the chemical injection rate, estimated consumption per well, and tank autonomy. It is written to be read by the client, designers, developers, and testers.

1.2 Scope

In scope	Out of scope (this release)
Multi-site registry of tanks, wells, pumps and products	Control or command of pumps/injection set points
Tank ↔ well association with rule validation	Orders, deliveries and invoicing
Ingestion of hourly tank volume and well production	Integration with ERP/CRM (only the real-time system is integrated)
Injection rate, estimated consumption per well and autonomy (days to 20%)	Native mobile apps (responsive web only)
Site dashboard, alerts, export, role-based access, audit trail	Chemical dosing recommendation/optimization

1.3 Client statements (source of requirements)

Each statement from the challenge brief received an ID so that every requirement can be traced back to it (see Section 11).

ID	Client statement (paraphrased)
S1	Basic characteristics of the equipment must be saved.
S2	Field technicians cannot install software on their computers.
S3	A tank can be associated with many wells; a well receives many chemical products.
S4	The system must serve several sites; users register equipment and associate them.
S5	Only Sales registers equipment; associations may be made by Sales and Field Technicians.
S6	The only integration is the real-time data system (hourly tank volume and individual well production).
S7	Show the injection rate of each tank and an estimated chemical consumption per well, based on well production, to support Sales.
S8	Tanks at a site have different capacities and construction materials.
S9	No more than 3 chemical products per well; products A and C cannot be applied to the same well.
S10	Sales forecasts tank operational autonomy (in days) until 20% of capacity is reached, to plan deliveries.

2. Users, Roles and Permissions

Role	Description	Key needs
Sales (Sales Representative / Manager)	Owns equipment master data; plans deliveries and potential sales.	Register equipment, forecast autonomy, see consumption per well, export data.

Role	Description	Key needs
Field Technician	Works at the site; knows physical piping and changes in the field.	Associate / dissociate tanks and wells quickly from a browser (tablet or laptop), see tank status.
System Administrator (assumed)	Manages users, site access, and configuration parameters.	User and site-access management, parameter configuration, audit.
Real-Time Data System (external)	Sends hourly tank volume and well production.	Reliable, idempotent data feed.

Permission matrix

Capability	Sales	Field Tech	Admin
Create/edit/deactivate sites, tanks, wells, pumps, products	Yes	No (read-only)	Yes*
Create / end tank–well associations	Yes	Yes	Yes
View dashboards, tank and well details	Yes	Yes	Yes
Export data	Yes	Yes (own sites)	Yes
Configure thresholds and rule parameters	No	No	Yes
Manage users and site access	No	No	Yes
View audit log	Yes (own sites)	No	Yes

*Assumption: Sales remains the only business role allowed to register equipment (S5); the Administrator role is technical and is expected to be restricted to a few named people. To be confirmed (OQ-08).

3. Assumptions and Open Questions

The brief is intentionally concise. The items below record interpretations made so that the specification is complete, and questions that must be validated with the client.

ID	Assumption / question	Working decision	Impact
OQ-01	Does a tank hold exactly one product? (Diagram shows one colour per tank.)	Yes – one product per tank.	BR-03, product count per well
OQ-02	If two tanks with the same product feed one well, does it count as 1 or 2 products for the limit of 3?	Counts as 1 distinct product.	BR-05
OQ-03	Diagram appears to show lines of products A and C reaching the same well. Is this an as-built error or an exception?	Treated as a rule violation to be flagged during data migration; the system blocks it.	BR-06, migration
OQ-04	Is a pump always 1:1 with a tank? Should pumps be registered?	1 pump per tank; pump registration is a should-have.	FR-05
OQ-05	Can a tank inject into wells of another site?	No – association only within the same site.	BR-07
OQ-06	Unit of well production (m ³ /day of oil? total fluid?) and unit of tank volume (L, m ³)?	Tank volume in liters; production in m ³ /day; unit conversion configurable.	BR-11, integration
OQ-07	Is 20% a fixed rule or configurable per tank/site?	Default 20%, configurable per tank by Sales.	BR-09
OQ-08	Who administers users and parameters?	A technical Administrator role.	Roles
OQ-09	How are refills reported? Does the real-time system flag deliveries?	Detect refills from volume jumps; may be replaced by an explicit event if available.	BR-10, FR-15

ID	Assumption / question	Working decision	Impact
OQ-10	Should historical associations be kept for consumption history?	Yes – associations have start and end dates.	BR-19
OQ-11	Language, time zones, data retention, and volume of sites/wells expected?	English UI; site time zone; hourly data kept for 2 years; ~100 sites / ~50 wells each.	NFR
OQ-12	Are there existing spreadsheets/registers to migrate?	Assumed yes; one-time CSV import (Could).	FR-27

4. Glossary

Term	Definition
Site	Physical production area with its wells, chemical tanks, flowlines, and oil tank (e.g., Site 1).
Tank (C-TK)	Chemical storage tank containing one chemical product; has capacity and construction material.
Well	Oil production well receiving chemical injection.
Pump	Equipment that moves chemical from a tank to the injection line.
Association	Link between a tank and a well, meaning that the tank's product is injected into the well.
Injection rate	Chemical volume leaving a tank per unit of time (L/h or L/day).
Estimated consumption (per well)	Share of the tank injection rate attributed to a well, proportional to its production.
Minimum level (threshold)	20% of tank capacity (default). Below it, the tank must be refilled.
Autonomy	Days until the tank volume reaches the minimum level at the current injection rate.
Refill event	Volume increase in a tank caused by a chemical delivery.
RTD	Real-Time Data system, the only external integration.

5. Business Rules

ID	Business rule	Source
BR-01	Only users with the Sales role can create, edit, or deactivate sites, tanks, wells, pumps, and products.	S5
BR-02	Users with the Sales or Field Technician role can create and end tank–well associations for the sites they can access.	S5
BR-03	A tank contains exactly one chemical product (product A, B, C, D...). The product is defined at tank registration.	S3, OQ-01
BR-04	A tank can be associated with many wells and a well with many tanks (many-to-many).	S3
BR-05	A well must never be associated with more than 3 distinct products . Two tanks of the same product count as one product. The system blocks the association that would exceed the limit.	S9
BR-06	Products A and C cannot be applied to the same well. This is stored as an incompatibility pair (product_a, product_b) so new pairs can be added without changes to the code. The system blocks the violating association and shows the conflicting tank and well.	S9
BR-07	Tanks and wells can only be associated when they belong to the same site.	S4, OQ-05
BR-08	Tank capacity is mandatory, greater than zero and expressed in liters; construction material is mandatory and selected from a list (e.g., carbon steel, stainless steel, HDPE, fiberglass).	S1, S8
BR-09	The minimum operational level of a tank is 20% of its capacity (configurable per tank, default 20%).	S10

ID	Business rule	Source
BR-13	Equipment codes are unique within a site (e.g., C-TK-1, Well-1, Pump-1).	S1, S4
BR-14	Equipment with history (readings or associations) cannot be deleted, only deactivated. Deactivating a tank or well ends all its active associations on that date.	S1
BR-15	Tank volume and well production come only from the RTD and cannot be edited by users.	S6
BR-16	Measurements are stored in UTC and displayed in the site time zone.	S4
BR-17	Every creation, change and association event stores user, role, date/time and previous value (audit).	S5
BR-18	Alert levels: Warning when autonomy ≤ 10 days; Critical when autonomy ≤ 5 days or volume $\leq$ minimum level (defaults, configurable per site).	S10
BR-19	Associations have a start date and an optional end date; consumption calculations use the associations valid at the measurement time.	S3, S7
BR-20	When an action is blocked by a rule, the message states the rule violated and the equipment involved.	S9

5.1 Calculation rules

ID	Rule	Formula/behaviour
BR-10	Tank injection rate. Computed from the hourly volume history in a rolling window (default 7 days, configurable).	Consumption = sum of hourly volume drops in the window ($V(t-1) - V(t)$ when positive). Hourly increases above the refill threshold (default 2% of capacity) are treated as refill events and excluded; smaller increases are treated as sensor noise and ignored. Rate (L/day) = Consumption $\div$ valid hours $\times 24$.
BR-11	Estimated consumption per well. The tank injection rate is distributed among the wells associated with the tank in proportion to their production.	Consumption(well, tank) = Rate(tank) $\times$ Production(well) $\div \Sigma$ Production(all active wells of that tank). If Σ Production = 0, the rate is shown as “unallocated”.
BR-12	Conservation. The sum of the estimated consumption of the wells of a tank must equal the tank injection rate (difference $\leq$ rounding).	Used as an automatic consistency check.
BR-21	Autonomy (days). Days until the tank reaches the minimum level.	Autonomy = $(V_current - MinLevel) \div Rate(L/day)$, where MinLevel = capacity $\times$ min_level_pct. If $V_current \leq MinLevel \rightarrow 0$ days, status Critical. If Rate = 0 $\rightarrow$ “not consuming”, autonomy not applicable.
BR-22	Data freshness. A reading older than 3 hours marks the tank as “stale”; calculations use the last valid reading and display a warning.	Freshness threshold configurable.
BR-23	Tank without associations. The rate is still calculated and displayed, flagged as “unassigned consumption”.	Helps detect missing registrations.

5.2 Worked example

Tank **C-TK-1** (Product A), capacity 10,000 L, current volume 4,200 L; volume 7 days ago 5,600 L with no refill. Minimum level = $20\% \times 10,000 = 2,000$ L.

Item	Calculation	Result
Injection rate	$(5,600 - 4,200) \div 7$	200 L/day (≈ 8.3 L/h)
Autonomy	$(4,200 - 2,000) \div 200$	11 days $\rightarrow$ status OK

Item	Calculation	Result
Associated wells and production	Well-1: 300 m ³ /d, Well-2: 500 m ³ /d, Well-4: 200 m ³ /d (total 1,000)	Shares 30% / 50% / 20%
Estimated consumption per well	200 × share	Well-1: 60 L/d · Well-2: 100 L/d · Well-4: 40 L/d (sum = 200 ✓)

5.3 Association validation examples

Situation	Result	Rule
Well-1 has products A and B; user adds tank with product C	Blocked: C conflicts with A already applied to Well-1	BR-06
Well-2 has products B, C, D; user adds tank with product A	Blocked: limit of 3 products (and A conflicts with C)	BR-05, BR-06
Well-3 has products A and B; user adds a second tank of product A	Allowed: A already counted (2 distinct products)	BR-05
Field Technician tries to create a new tank	Blocked: only Sales can register equipment	BR-01
Tank of Site 1 associated with a well of Site 2	Blocked: different sites	BR-07

6. Functional Requirements

Priority uses MoSCoW: **M** = Must, **S** = Should, **C** = Could.

ID	Requirement	Priority	Rules	Source
FR-01	Manage sites (create, edit, deactivate) with name, code, and time zone.	M	BR-01, BR-16	S4
FR-02	Manage the chemical product catalog (A, B, C, D...) and the incompatibility pairs.	M	BR-01, BR-06	S9
FR-03	Register tanks: code, site, product, capacity (L), construction material, minimum level %, status.	M	BR-01, 03, 08, 13	S1, S5, S8
FR-04	Register wells: code, name, site, status.	M	BR-01, BR-13	S1, S5
FR-05	Register pumps linked to a tank: code, nominal flow, status.	S	BR-01, BR-13	S1
FR-06	Edit and deactivate equipment; deletion is not allowed when there is history.	M	BR-14	S1
FR-07	List, search, and filter equipment by site, type, product, material, and status.	M	–	S4
FR-08	Associate a tank to one or more wells (and a well to tanks) with a start date.	M	BR-02, 04, 07, 19	S3, S5
FR-09	Validate the limit of 3 distinct products per well before saving the association.	M	BR-05, BR-20	S9
FR-10	Validate incompatible products (A × C) before saving the association.	M	BR-06, BR-20	S9
FR-11	End an association keeping its history.	M	BR-19	S3
FR-12	Visualize associations as a matrix (tanks × wells) and per equipment, highlighting rule conflicts.	S	BR-05, 06	S3, S4
FR-13	Receive hourly tank volume and well production from the RTD; idempotent, reprocessable.	M	BR-15, BR-16	S6

ID	Requirement	Priority	Rules	Source
FR-14	Validate and flag incoming data (missing, negative, above capacity, unknown tag, stale).	M	BR-22	S6
FR-15	Detect and log refill events from volume increases.	S	BR-10	S7, S10
FR-16	Calculate and display the injection rate per tank (L/h and L/day).	M	BR-10	S7
FR-17	Calculate and display estimated consumption per well and per tank–well pair.	M	BR-11, 12, 19	S7
FR-18	Calculate and display tank autonomy in days until the minimum level.	M	BR-09, BR-21	S10
FR-19	Site dashboard: tanks sorted by autonomy, status colours, alerts, data freshness.	M	BR-18, 21, 22	S7, S10
FR-20	Tank detail: volume trend with 20% line, rate, autonomy, associated wells, refill events.	M	BR-10, 21	S7
FR-21	Well detail: associated tanks/products, production, estimated consumption per product.	S	BR-11	S7
FR-22	Alerts (in-app and e-mail) for warning/critical autonomy, stale data and rule violations in imported data.	S	BR-18, 22	S10
FR-23	Export dashboards and lists to CSV / XLSX / PDF.	S	–	S7
FR-24	Authentication and role-based access control, with access limited by site.	M	BR-01, 02	S5
FR-25	Audit log of registrations and associations, searchable by user, equipment and date.	S	BR-17	S5
FR-26	Configure parameters (minimum level %, rate window, refill threshold, alert levels, incompatibility pairs).	S	BR-06, 09, 10, 18	S10
FR-27	One-time / periodic CSV import of equipment for onboarding new sites.	C	BR-01, 13	S4

7. Non-Functional Requirements

ID	Category	Requirement	Priority
NFR-01	Deployment / Access	Browser-based; no client installation or plug-ins. Supports current and previous major versions of Chrome, Edge, Firefox, and Safari.	M
NFR-02	Usability	Responsive from tablet size (10") to desktop; key field tasks (association) achievable in ≤ 3 clicks from the site dashboard; user can learn the association flow in < 15 minutes without training.	M
NFR-03	Performance	Dashboard loads in ≤ 3 s (p95) for a site with 50 wells and 60 tanks; association validation responds in ≤ 1 s.	M
NFR-04	Data timeliness	Hourly data visible in the system ≤ 15 minutes after being received from the RTD.	M
NFR-05	Scalability	Supports at least 100 sites, 5,000 wells, 3,000 tanks, and 2 years of hourly history without degradation.	S
NFR-06	Availability	99.5% monthly availability in business hours; RTD outage does not prevent registration and consultation.	M
NFR-07	Security	TLS 1.2+, corporate SSO or username/password with MFA, RBAC, least privilege, site-level data segregation, OWASP Top 10 mitigations.	M
NFR-08	Integration reliability	Ingestion is idempotent, retries with back-off, and stores raw messages for reprocessing; failures raise alerts.	M
NFR-09	Auditability	All changes are recorded with user, role, timestamp, old and new value; logs retained ≥ 2 years.	S

ID	Category	Requirement	Priority
NFR-10	Data retention & backup	Hourly data retained for 2 years, then aggregated daily; daily backups, RPO ≤ 24 h, RTO ≤ 8 h.	S
NFR-11	Accuracy	Calculations reproduce the specification examples exactly (tolerance ±0.1 L/day) and are covered by automated tests.	M
NFR-12	Localization	English UI; units configurable (L / m ³ , m ³ /day); dates in site time zone; internationalization-ready texts.	S
NFR-13	Accessibility	WCAG 2.1 AA for main flows; status not communicated by colour alone.	S
NFR-14	Maintainability & observability	Configurable parameters without code change; centralized logs and health metrics.	S

8. Data Model

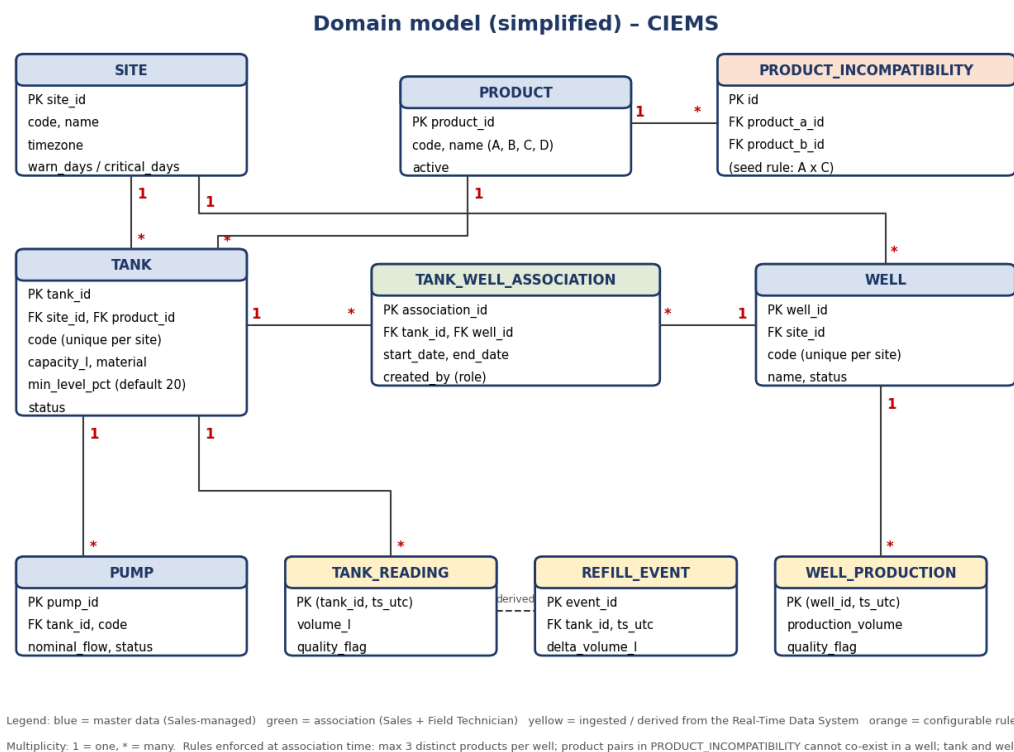


Figure 1 – Simplified domain model. Association is a first-class entity with start/end dates (BR-19). Incompatibility is data, not code (BR-06).

Integration summary (RTD → CIEMS)

Item	Specification (to be detailed in D11)
Direction/frequency	RTD → CIEMS, hourly, read-only (BR-15)
Tank message	tank_tag, timestamp (UTC), volume (L), quality flag
Well message	well_tag, timestamp (UTC), production value and unit, quality flag
Mapping	Equipment code ↔ RTD tag stored in the equipment record (Sales-maintained)
Error handling	Unknown tag → quarantine and alert; duplicate timestamp → ignored (idempotent); gap → flagged missing, no interpolation of volumes

9. User Stories

Acceptance criteria use the Given / When / Then format. Estimates are left to the development team.

ID	Story	Acceptance criteria	Prio	Refs
US-01	As a Sales user , I want to register a chemical tank with product, capacity, and material, so that it can be associated and monitored.	Given I am a Sales user, when I save a tank with all mandatory fields, then it is listed on the site with status Active. Given the code already exists in the site, then saving is blocked with a clear message. Given capacity ≤ 0 or no material, then saving is blocked.	M	FR-03; BR-01, 03, 08, 13
US-02	As a Sales user , I want to register wells, so that tanks can be associated with them.	Given I am Sales, when I save a well with code, name, and site, then it appears in the well list. Given I am a Field Technician, then the “New well” option is not available.	M	FR-04; BR-01, 13
US-03	As a Sales user , I want to register pumps linked to tanks, so that the physical injection setup is complete.	Given a tank exists, when I save a pump with a code and tank, then it is shown in the tank detail.	S	FR-05
US-04	As a Sales user , I want to edit or deactivate equipment, so that the register reflects the field.	Given a tank has readings, when I try to delete it, then only “Deactivate” is offered. Given I deactivate a tank, then its active associations are ended on that date.	M	FR-06; BR-14
US-05	As a Sales user or Field Technician , I want to associate a tank to one or more wells, so that the injection points are known.	Given a tank and wells of the same site, when I confirm, then associations are created with today as the start date. Given different sites, then the action is blocked.	M	FR-08; BR-02, 04, 07
US-06	As a user , I want the system to stop associations that exceed 3 products in a well.	Given Well-2 has products B, C, and D, when I add product A, then the system blocks it and shows the 3 products already applied. Given the well has A and B, and I add another tank of A, then it is allowed.	M	FR-09; BR-05
US-07	As a user , I want the system to stop products A and C in the same well.	Given Well-1 receives A, when I add a tank with C, then it is blocked, and the message names the tank and the rule.	M	FR-10; BR-06, 20
US-08	As a Field Technician , I want to end an association when a line is disconnected, keeping history.	Given an active association, when I end it with a date, then it no longer counts for the limit of 3 products and history remains visible.	M	FR-11; BR-19
US-09	As a Sales user , I want to see each tank’s autonomy in days, so that I can plan deliveries.	Given hourly readings, when I open the dashboard, then tanks are sorted by autonomy in ascending order with status colours. Given volume $\leq$ minimum level, then autonomy is 0 and status is Critical.	M	FR-18, 19; BR-09, 21
US-10	As a Sales user , I want to see the injection rate of each tank, so that I understand real consumption.	Given the example in 5.2, then rate = 200 L/day. Given a refill in the window, then the refill is excluded from consumption.	M	FR-16; BR-10
US-11	As a Sales user , I want an estimated consumption per well, so that I can support sales and pricing conversations.	Given the example in 5.2, the wells show 60, 100, and 40 L/day, and the sum equals the tank rate.	M	FR-17; BR-11, 12
US-12	As a Sales user , I want to see the tank volume trend against the 20% line, so that I can explain the forecast to the client.	Given a tank, when I open its details, then the last 30 days are plotted with refill markers and the minimum level.	M	FR-20
US-13	As a Sales user , I want alerts when a tank is close to the minimum level, so that deliveries are planned in time.	Given autonomy ≤ 10 days, then a warning is raised once; given ≤ 5 days, a critical alert is raised.	S	FR-22; BR-18

ID	Story	Acceptance criteria	Prio	Refs
US-14	As a user , I want to know when data is stale, so that I do not trust an outdated forecast.	Given no reading for over 3 hours, then the tank shows “stale” with the time of the last reading.	M	FR-14; BR-22
US-15	As a Sales user , I want to export the dashboard, so that I can share it with clients and managers.	Given filters are applied, when I export, then a CSV/XLSX/PDF with the same data is generated.	S	FR-23
US-16	As an Administrator , I want to manage users and site access, so that each person sees only their sites.	Given a user assigned to Site 1, then Site 2 data is not visible or accessible by URL.	M	FR-24
US-17	As a Sales manager , I want an audit trail, so that I know who changed a registration or association.	Given an association was ended, then the log shows who, when, and previous values.	S	FR-25; BR-17
US-18	As an Administrator , I want to configure parameters and incompatibility pairs, so that rules change without new software versions.	Given a new incompatibility pair is saved, then it is enforced in new associations immediately.	S	FR-26

10. Use Cases

UC-01 – Associate a tank to wells

Field	Description
Actors	Sales, Field Technician
Preconditions	User is authenticated with access to the site; tank and wells are active.
Trigger	User opens the Association Manager (SCR-04).
Main flow	1. User selects the site. 2. User selects a tank. 3. System lists the wells of the site with their current products. 4. User ticks one or more wells and clicks “Save”. 5. System validates BR-05, BR-06 and BR-07 for each well. 6. System creates the associations with a start date and logs the audit event. 7. The system confirms and refreshes the matrix and estimated consumption.
Alternative flows	5a. A rule fails: the system blocks only the invalid well, explains the rule and the conflicting tank, and allows the user to save the other valid wells (or cancel). 4a. User selects a date in the past: allowed only for Sales (assumed).
Postconditions	Associations exist and are used in the next calculation cycle.
Rules/requirements	BR-02, 04, 05, 06, 07, 19, 20 · FR-08, 09, 10, 12 · US-05, 06, 07

UC-02 – Monitor tank autonomy

Field	Description
Actors	Sales (primary), Field Technician
Preconditions	Hourly data received for the site.
Main flow	1. User opens the site dashboard (SCR-01). 2. System shows tanks sorted by autonomy, current volume %, rate, status, and data freshness. 3. User filters by product or status. 4. User opens a tank (SCR-05) to see the trend, refill events, and estimated consumption per well. 5. User exports the view if needed.

Field	Description
Alternative flows	2a. Data stale: tank marked “stale”, last valid values shown. 2b. Tank not consuming: autonomy shows “n/a”. 2c. No associations: “unassigned consumption” warning.
Postconditions	None (read-only).
Rules/requirements	BR-09, 10, 11, 18, 21, 22, 23 · FR-16 to 20, 22, 23 · US-09 to 14

UC-03 – Register a chemical tank

Field	Description
Actors	Sales
Preconditions	Site and product exist.
Main flow	1. Sales opens Equipment > Tanks and selects “New tank” (SCR-02). 2. Sales fills in code, site, product, capacity, material, minimum level %, RTD tag (SCR-03). 3. System validates mandatory fields and uniqueness (BR-08, BR-13). 4. System saves the tank and logs the event.
Alternative flows	3a. Validation error: fields are highlighted with messages. 1a. A Field Technician opens the list: “New” is hidden (BR-01).
Postconditions	Tank is available for associations and for data mapping.
Rules/requirements	BR-01, 03, 08, 13, 17 · FR-03, 06, 07 · US-01, 04

11. Requirements Traceability Matrix

Prototype screens (file 04): SCR-01 Site Dashboard · SCR-02 Equipment List · SCR-03 Tank Form · SCR-04 Association Manager · SCR-05 Tank Detail · SCR-06 Well Detail · SCR-07 Alerts & Reports.

Client statement	Functional requirements	Business rules	User stories	Screens	NFR	Test focus (UAT)
S1 Save basic equipment data	FR-03, 04, 05, 06, 07	BR-08, 13, 14	US-01–04	SCR-02, 03	NFR-09	Mandatory fields, uniqueness, deactivation
S2 No software installation	All	–	All	All	NFR-01, 02	Run key flows in supported browsers
S3 Many-to-many tank/well	FR-08, 11, 12	BR-04, 19	US-05, 08	SCR-04, 06	NFR-03	Multiple associations, history
S4 Multi-site	FR-01, 07, 24, 27	BR-07, 13, 16	US-05, 16	SCR-01, 02	NFR-05, 07	Site isolation and cross-site block
S5 Sales registers; Sales+FT associate	FR-24, 25	BR-01, 02, 17	US-02, 05, 16, 17	SCR-02, 04	NFR-07	Permission matrix
S6 Only RTD integration	FR-13, 14, 15	BR-15, 16, 22	US-14	SCR-01, 07	NFR-04, 06, 08	Late, duplicate, missing data
S7 Injection rate & consumption per well	FR-16, 17, 20, 21, 23	BR-10, 11, 12, 23	US-10, 11, 12, 15	SCR-05, 06	NFR-03, 11	Worked example 5.2, refill exclusion
S8 Different capacities/materials	FR-03	BR-08	US-01	SCR-03	–	Capacity units and material list
S9 Max 3 products; A × C	FR-09, 10, 02, 26	BR-05, 06, 20	US-06, 07, 18	SCR-04	NFR-03	Examples in 5.3
S10 Autonomy to 20%	FR-18, 19, 22	BR-09, 18, 21	US-09, 13	SCR-01, 05, 07	NFR-04, 11	Autonomy formula, thresholds, alerts